

Operating Manual - English

RDX USB 3.0 QuikStor Drive

Operating Manual

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For your safety

This manual contains important information for safely and correctly using this product.

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This product is a "Class A" ITE (Information Technology Equipment). In a domestic environment this product may cause radio interference, in which case the user may be required to take appropriate measures.

VCCI-A

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The aluminum electrolytic capacitors used in the printed circuit board assemblies of the product and in the mouse and keyboard are limited-life components. Use of these components beyond their operating life may result in electrolyte leakage or depletion, potentially causing emission of foul odor or smoke.

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High safety use

This product has been designed and manufactured to be used in commercial and/or industrial areas as a server.

The product is not suitable for use at visual display workplaces according to §2 of the Workplace Regulations (applies to all server systems except TX server systems).

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The device has not been designed or manufactured for uses which demand an extremely high level of safety and carry a direct and serious risk of life or body if such safety cannot be assured.

These uses include control of nuclear reactions in nuclear power plants, automatic airplane flight control, air traffic control, traffic control in mass transport systems, medical devices for life support, and missile guidance control in weapons systems (hereafter, "high safety use"). Customers should not use this product for high safety use unless measures are in place for ensuring the level of safety demanded of such use. Please consult the sales staff of Fsas Technologies if intending to use this product for high safety use.

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This product may be affected by a momentary voltage drop in the power supply caused by lightning. To prevent a momentary voltage drop, the use of an uninterruptible power supply is recommended.

(This notice follows the guidelines of Voltage Dip Immunity of Personal Computer issued by JEITA, the Japan Electronics and Information Technology Industries Association.)

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Harmonic Current Standards

This product conforms to harmonic current standard JIS C 61000-3-2.

Only for Japan: About SATA HDDs

The SATA version of this server supports HDDs with SATA/BC-SATA storage interfaces. Please note that the usage and operation conditions differ depending on the type of HDD used.

For more information on the usage and operation conditions of each available type of HDD, see the following internet address:

<https://www.fsastech.com/ja-jp/products/primergy/peripheral/disk/>

Only for Japan:

Shielded LAN cables should be used in this product.

UK Importer information

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Version history

Issue number	Issue date	Description
V 1.0	10/2012	Initial release
V 2.0	12/2017	Appendix added
V 3.0	07/2023	Minor editorial changes Following sections or chapters were updated: • Installation instructions • RDX Utility • Troubleshooting
V 4.0	10/2023	Following sections were updated: • Installation instructions • Overview (RDX Utility) Following chapters were added: • RDX Manager • Limitation
V 5.0	03/2026	Company name changed to Fsas Technologies

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1 Introduction

1.1 Overview

The RDX drive is a removable, ruggedized, hard disk drive system. It features backup application compatibility, high speed transfer rate for fast backups, portable cartridges and high level of data reliability.

The RDX drive is available as internal USB device.



Figure 1: Internal USB device

1 3.5-inch internal USB device

2 5.25-inch internal USB device

1.2 Supported media

The RDX hard drive system supports any RDX cartridge which Fsas Technologies recommends. RDX cartridges are available in different capacity configurations depending on your storage requirements.

1.3 Software

The RDX drive works with all industry backup applications that support removable disk media. See your backup software manual for instructions on backing up data.

For Windows there are 2 tools that can be used:

- RDX Utility, see ["RDX Utility" on page 25](#).
- RDX Manager, see ["RDX Manager" on page 27](#).

1.4 Media equipment

RDX cartridges must be ordered separately! All former media are compatible also with old and new RDX drives.



An RDX Media Cartridge contains a 2.5-inch hard disk drive. It must be handled with care to ensure data integrity and long life.

1.5 Front panel features

The front of the RDX unit has the following features and LEDs:

- Eject Button / Power LED
- Activity LED (located on cartridge)
- Emergency Eject Hole



Figure 2: Front panel features

- | | | | |
|---|----------------------|---|--------------------------|
| 1 | Emergency Eject Hole | 3 | Eject Button / Power LED |
| 2 | Activity LED | | |

The eject button is illuminated by a power LED. The operating states of the power LED are described in ["Power LED" on page 13](#). The Activity LED displays the current status of the cartridge as described in ["Activity LED" on page 14](#).

Power LED

LED Status	Meaning	Description
OFF	No Power	Unit does not have power.
Steady GREEN	Ready	Power is on and the RDX unit is working properly.
Blinking GREEN	Ejecting	Unit is ejecting the cartridge.

Introduction

LED Status	Meaning	Description
Steady AMBER	Warning	RDX unit has detected a warning condition.
Blinking AMBER	Prevent (eject action)	Host computer is accessing the media, and the user has pressed the eject button. The unit will blink amber for a few seconds then return to a steady green. The unit will eject the cartridge after the host access completes.

Activity LED

LED Status	Meaning	Description
OFF	No Power	Cartridge is not inserted properly or the unit has no power.
Steady GREEN	Ready	Cartridge is ready.
Blinking GREEN	Accessing	Unit is reading, writing, seeking.
Steady AMBER	Warning	RDX unit has detected a warning condition.

2 Installing the RDX unit

2.1 Before you begin

The RDX unit ships ready to install in a standard 3.5-inch or 5.25-inch half-height drive bay depending on the configuration you order.

2.2 Important Information on electro-static damage

To protect your computer and drive from electro-static damage (ESD), follow these guidelines:

- ▶ Before you open your computer or remove the RDX unit from its antistatic bag, ground yourself by touching the metal chassis of your computer or some other grounded object. For optimal protection, use an electrostatic grounding strap.



Your computer must be plugged into a grounded electrical outlet to be properly grounded.

- ▶ Do not touch any printed circuit boards or electronic chips on your RDX device or in your computer.
- ▶ Turn your computer and any peripheral devices off before installing the RDX unit.

2.3 Preparing for installation

Dependent on the type of server, it is required to attach mounting rails to the RDX drive. For further information, please refer to your server's "Upgrade and Maintenance Manual".

2.4 Installation instructions

Proceed as follows to mount an RDX QuikStor drive:

- ▶ Carefully unpack the drive and accessories.
- ▶ Switch off the computer.
- ▶ Remove AC plug.
- ▶ Open the computer cover, using a screwdriver if necessary.
- ▶ Slide the RDX drive into an unused drive bay until the bezel is flush with the front of the computer.
- ▶ Mount the drive as appropriate for your server.



Do not over-tighten the screws.

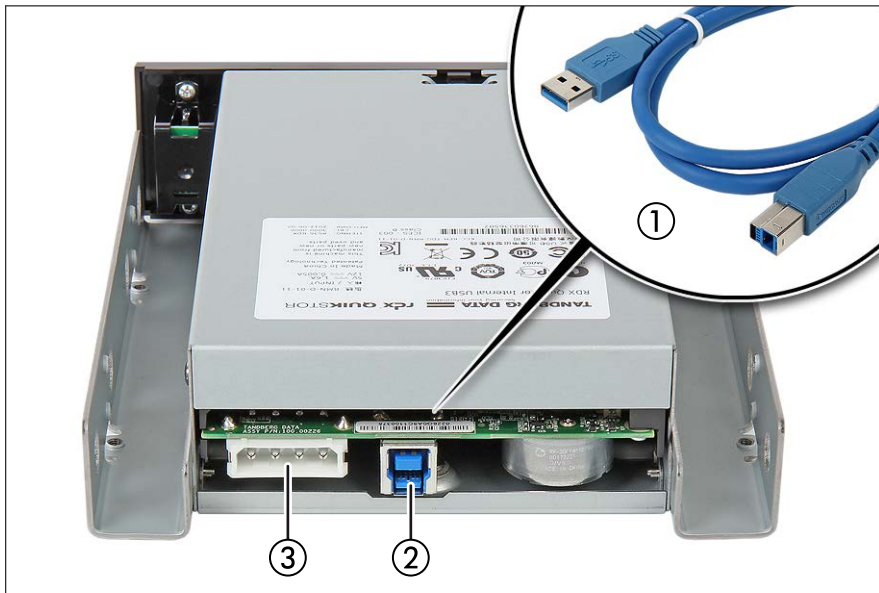


Figure 3: USB connector on RDX drive connector panel

- | | | | |
|---|--------------------------|---|-----------------|
| 1 | USB cable | 3 | Power connector |
| 2 | Type B USB 3.0 connector | | |

- ▶ Connect the USB cable type B (1) to the RDX drive (2) and connect the end of the USB cable to the USB 3.0 PCI card (see ["USB connector on USB expansion card" on page 17](#)) or the appropriate USB 3.0 connector on the system board (see your server's Upgrade and Maintenance Manual).



To ensure best performance, use USB 3.0 host with appropriate USB 3.0 cable.

- ▶ Connect the power cable to the RDX drive (3).

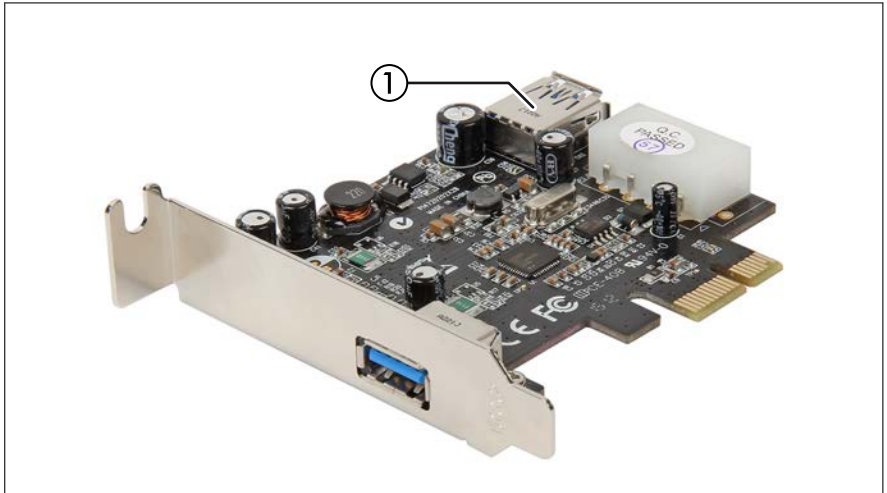


Figure 4: USB connector on USB expansion card

- 1 Type A internal USB 3.0 connector

- ▶ Check all cables in your computer system. Before you reinstall your computer cover, ensure that all cables are correctly routed so that they will not bind or become crimped. Also, make sure that no cables have loosened during the installation.

3 Using the RDX unit

3.1 Operating the RDX unit

This section describes typical activities involved in operating and maintaining the RDX unit. For instructions on using your backup software, see the manual that was delivered with your software.

3.2 Handling RDX cartridges

Although RDX cartridges are ruggedly built, they must be handled with care to ensure data integrity and long life. Observe the following precautions:

- ▶ Store cartridges in their protective cases when you are not using them.
- ▶ Keep cartridges away from dust, dirt and moisture.
- ▶ Avoid direct sunlight, heat sources and temperature extremes or rapid changes in temperature.
- ▶ Never try to force the cartridge into the unit. If it is difficult to insert, check the orientation and try again.
- ▶ Do not drop cartridges where non-operational specifications may be exceeded.

3.3 Write protection

Write protection should be enabled whenever you wish to protect the RDX cartridge from accidental overwriting or deletion. When the RDX cartridge is write-protected, the system can read data, but can neither overwrite nor delete data.



Figure 5: Toggling write protection

- Slide the red write-protection switch (see close-up) to the right or left into the desired position.



When the switch is set to the write-protection position, no data can be written to the cartridge.

When the switch is in the unlocked position, data can be written to the cartridge.

3.4 Loading and unloading cartridges

The front of your unit has an opening for inserting RDX cartridges. A protective door covers the opening. Below and to the right of the door is the power LED that doubles as a cartridge-eject button.

3.4.1 To load a cartridge

i The cartridge only fits in one orientation.

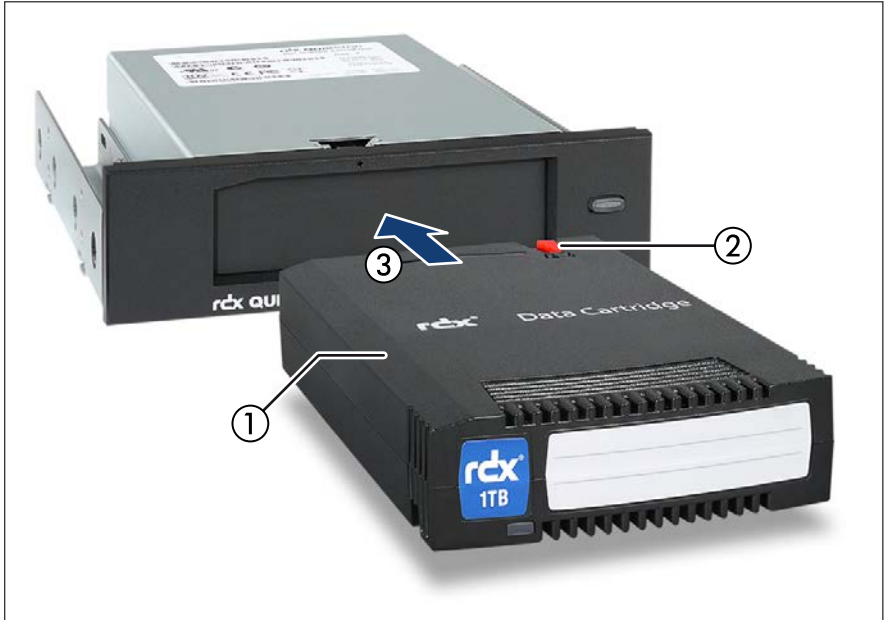


Figure 6: Loading the cartridge

- ▶ Insert the cartridge in the unit with the keyed corner (1) facing the upper left corner of the unit and the write-protect tab (2) on the back right.
- ▶ Push the cartridge gently into the unit until it is locked in place with a clicking sound (3).

i If the cartridge is not fully inserted, it may not work properly and cause writing or reading errors.

3.4.2 To unload a cartridge

To unload a cartridge using the mouse right-click method (Windows only)

- ▶ While running Windows Explorer locate the drive letter for the device.
- ▶ Right-click on the device and select **Eject**.
- ▶ When the cartridge has been ejected by the unit, grasp the exposed sides of the cartridge and pull the cartridge straight out.

To unload a cartridge using the eject button



Figure 7: Unloading the cartridge

- ▶ Push the eject button (1).
- ▶ When the cartridge has been ejected by the unit, grasp the exposed sides of the cartridge and pull the cartridge straight out (2).

3.4.3 To emergency eject a cartridge

i Do not emergency eject a RDX cartridge while the cartridge activity LED is flashing.

If the cartridge does not eject, using either the eject button or the right-click eject methods then use the following emergency eject hole method to remove the cartridge.



Figure 8: Emergency eject of the cartridge

- ▶ Insert a straightened paper clip into the Emergency Eject Hole (1).
- ▶ When the cartridge has been ejected by the unit, grasp the exposed sides of the cartridge and pull the cartridge straight out (2).

4 RDX Utility

4.1 Overview

The **RDX Utility** is a Windows application for Windows Server 2019 or older versions that allows you to diagnose problems with your RDX device. With the utility, you can:

- Identify RDX devices attached to your computer.
- Retrieve device and cartridge status information.

4.2 Using the RDX Utility

To run the **RDX Utility** for Windows:

- ▶ Click on **Start**.
- ▶ Click on **All Programs**.
- ▶ Click on **RDX Utility**.
- ▶ For instructions on using the **RDX Utility**, click on **Help**.

5 RDX Manager

5.1 Overview

The **RDX Manager** is a Windows application for Windows Server 2022 and Windows Server 2019 that allows you to diagnose problems with your RDX device. With the manager, you can:

- Identify RDX devices attached to your computer.
- Retrieve device and cartridge status information.
- Partition and format the cartridge.
- Change drive mode.
- Test the unit and the cartridge.

5.2 Using the RDX Manager

To run the **RDX Manager** for Windows:

- ▶ Click on **Start**.
- ▶ Click on **All Programs**.
- ▶ Click on **RDX Manager**.
- ▶ For instructions on using the **RDX Manager**, click on **Help**.

5.3 RDX Manager Drive List

The **RDX Manager Drive List** allows you to manage your RDX devices in one place.

A window is displayed with a table listing all discoverable RDX devices.

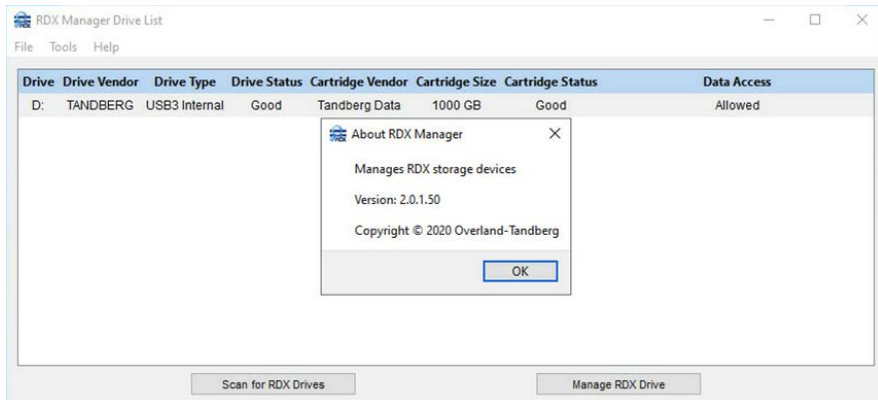


Figure 9: RDX Manager Drive List

The main window of the **RDX Manager Drive List** contains the following sections:

- The title bar resides at the top of the main window and contains the following menus:
 - **Files**
 - **Tools**
 - **Help**
- The working area resides below of the title bar that contains the following:
 - The table showing information about the installed RDX devices and any inserted cartridges.
 - Two buttons at the bottom of the window.

Table listing installed RDX drives and inserted cartridges

Drive

The drive letter used by the RDX device.

Drive Vendor

Manufacturer of the unit.

Drive Type

Connection type.

Drive Status

Current status of the drive.

Cartridge Vendor

Manufacturer of the cartridge.

Cartridge Size

Capacity of a cartridge (if inserted in the device).

Cartridge Status

Current status of the cartridge (if inserted in the device).

Data Access

Indicates if data access is allowed.

Buttons**Scan for RDX Drives**

Scans all discoverable RDX devices.

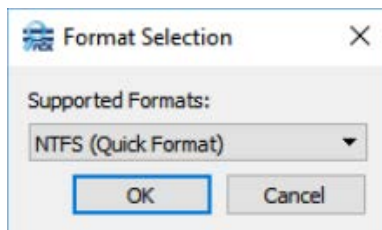
Manage RDX Drive

Manages your RDX devices.

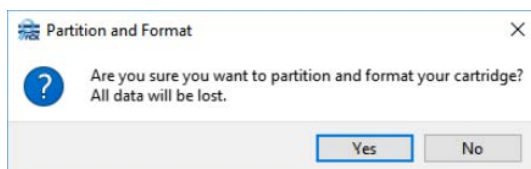
5.4 Partition and Format Cartridge

The new RDX cartridges are formatted using the NTFS file system. You can partition and format your cartridge based on your operating system.

- ▶ Click the **Partition and Format Cartridge** button on the Windows computer, and a **Format Selection** dialog will be open.
- ▶ Select either **NTFS (Quick Format)** or **exFAT (Quick Format)** option and click **OK**.



- ▶ Click **Yes** to start the format.

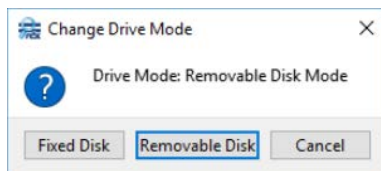


Progress and result are shown in the bottom message area.

5.5 Change Drive Mode

This utility enables changing between **Fixed Disk** and **Removable Disk**.

- ▶ When you click **Change Drive Mode**, a dialog box appears showing the current drive mode (with that mode button highlighted in blue).
- ▶ Click the new mode (**Fixed Disk** or **Removable Disk**) to activate it. Click **Cancel** to close the function without making any changes.



The change completes quickly. Progress and result are shown in the bottom message area.

6 Software installation

6.1 Windows

It is not required to install a driver under Windows OS. Windows recognizes the drive as a removable disk and installs its native driver.

6.2 Linux

For Linux based operating systems (Novell SUSE, RedHat), it is not required to install a disk driver. The disk drive can be accessed by using device handles. At the terminal console type "dmesg" and scroll through the message to verify that the disk drive has been detected by the USB controller.

For example, you can access the disk drive over the handles `/media/usbdisk` or with the commands `mt`, `tar`, `cpio` and `dd`. Type `man` command at the console to get detailed information.

7 Troubleshooting

7.1 The power LED is OFF

If the power LED is off, the RDX internal unit does not have power. Follow these steps to verify the cable connections in the rear of the unit:

- ▶ Turn off the computer.
- ▶ Check the USB power and USB data cable connections on the rear of the unit.
- ▶ Turn on the computer.

7.2 The power LED is steady AMBER

If the power LED is a steady amber, the RDX unit is not working properly. This may be due to defective hardware.

- ▶ If the problem persists after turning the server off and on again, please contact your support partner.
- ▶ For further instructions on how to remove a stuck cartridge, see ["To emergency eject a cartridge" on page 23](#).

7.3 The power LED is blinking AMBER, and the cartridge does not eject

If the power LED is blinking amber after you press the eject button, the Host computer is accessing the media. The RDX unit will eject the cartridge after the Host access completes.

If the cartridge does not eject after the Host computer access completes, you may not be running the correct unit software.

For further instructions, see ["To unload a cartridge" on page 22](#).

7.4 The cartridge LED is steady AMBER

The cartridge LED is located on the front of the cartridge. If this LED is amber, the cartridge is not working properly. The error is due to one of the following conditions:

- The cartridge is not compatible with the RDX unit.
- The cartridge has been severely damaged.
- The cartridge has not been fully inserted.

7.5 The cartridge does not eject

For instructions on how to remove a stuck cartridge, see ["To emergency eject a cartridge" on page 23](#).

7.6 The unit does not appear in the device manager (Windows)

If the unit does not appear in the Windows **Device Manager** and the power LED is green, follow these steps to rescan the hardware:

- ▶ Ensure that all cables have been properly installed.

Bring up the **Device Manager**:

- ▶ Right-click on **My Computer**.
- ▶ Select **Manage**.
- ▶ Select **Device Manager**.
- ▶ Right-click on the top folder (this should be your computer's name).
- ▶ Select **Scan for Hardware Changes**.

You should see your RDX device listed under **Disk drives**. If the device does not appear, restart the computer.

7.7 Unit performance seems slow

For Linux users

Systems running Linux may see reduced performance when doing backups on RDX cartridges configured for FAT32. For best performance configure the file system on the RDX cartridges for ext4 or XFS.

Checking cable connections

For best performance, ensure to use an appropriate USB 3.0 cable.

7.8 A write error occurs

If the backup application reports a write error, follow these steps:

- ▶ Verify that the write protect switch on the cartridge is set to the “unlock position”, see ["Toggling write protection" on page 20](#).
- ▶ Check the activity LED on the front of the cartridge.

If this activity LED is amber, the cartridge is not working properly. This may be due to severe damage to the cartridge.

- ▶ Replace the cartridge if the error persists.

8 Limitation

8.1 Formatting the disk in Fixed Disk mode is not possible

When your cartridge's drive mode is set to **Fixed Disk**, the **RDX Utility** or the **RDX Manager** cannot format the disk.

In this case, please format the cartridge in the disk management of Windows.

8.2 Updating the RDX device firmware is not possible

Fsas Technologies currently has no plans for updating the default RDX device's firmware version.



This function is currently not supported.

8.3 Windows cannot format files in FAT32 that are larger than 4 GB

The FAT32 file system has a built-in limitation on the size of the files that it may contain. The size of each individual file may not exceed 4 GB.

If the RDX cartridge is formatted in FAT32 and the file size is larger than 4 GB, Windows does not allow the user to reformat the cartridge as FAT32.

Windows Explorer may not display a volume label for the inserted cartridge the first time.

8.4 An error message appears when accessing an RDX cartridge

The error message ID 153 appears in the Windows system event log if the RDX cartridge is accessed first after Windows starts.

8.5 Encryption disk is not possible

Based on the default RDX device's firmware version, encryption disk is not supported. Fsas Technologies currently has no plans for updating the default RDX device's firmware version.



This function is currently not supported.

8.6 Change SATA III mode is not possible

Fsas Technologies does not support SSD cartridges.



This function is currently not supported.

9 Appendix

9.1 China - Restriction of Hazardous Substance Table (RoHS)

产品中有害物质的名称及含量

部件名称	有害物质					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr (VI))	多溴联苯 (PBB)	多溴二苯醚 (PBDE)
机箱/风扇	X	O	O	O	O	O
印刷电路板组件	X	O	O	O	O	O

本表格依据SJ/T 11364 的规定编制。

O: 表示该有害物质在该部件所有均质材料中的含量均在 GB/T 26572 规定的限量要求以下。

X: 表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T 26572 规定的限量要求。

9.2 Taiwan - BSMI Restriction of Hazardous Substance Table (RoHS)

本產品符合臺灣RoHS標準CNS15663

單元	限用物質及其化學符號					
	鉛 (Pb)	汞 (Hg)	鎘 (Cd)	六價鉻 (Cr ⁺⁶)	多溴聯苯 (PBB)	多溴二苯醚 (PBDE)
外殼	O	O	O	O	O	O
電路板組件	—	O	O	O	O	O
電線/連接器	O	O	O	O	O	O
配件	O	O	O	O	O	O

備考1. “O” 係指該項限用物質之百分比含量未超出百分比含量基準值。

備考2. “—” 係指該項限用物質為排除項目。